Task 5: Exploratory Data Analysis (EDA)

Objective

Extract insights using visual and statistical exploration with Python (Pandas, Matplotlib, Seaborn).

Dataset

Titanic Dataset (train.csv).

Data Cleaning

• Checked missing values using .isnull().sum().
• Filled missing Age values using the median.
• Filled missing Embarked values using the mode.
• Dropped Cabin because it contained many missing values.
• Converted Age from float to integer after cleaning.

Statistical Exploration

• Used .info() to inspect the dataset.
• Used .describe() for summary statistics.
• Used .value_counts() for Survived, Sex, Pclass, and Embarked.

Visualizations

• Histograms
• Boxplots
• Scatterplot
• Pairplot
• Correlation Heatmap

Key Observations

• Most passengers were between 20 and 40 years old.
• Most passengers belonged to third class.
• Male passengers were more than female passengers.
• Female passengers had a higher survival rate.
• First-class passengers survived more often than third-class passengers.
• Fare contains several outliers.

Summary of Findings

The exploratory data analysis identified important patterns in the Titanic dataset. Passenger class and gender strongly influenced survival outcomes. Most passengers were young adults, and ticket fares showed a wide variation with several outliers. The analysis successfully revealed trends, relationships, and anomalies in the data.

Outcome

Gained practical experience in finding patterns, trends, relationships, and anomalies using EDA techniques.